

AVINASH D

Python Backend Engineer — AI / ML Engineer

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Professional Summary

Python Backend and AI/ML Engineer with 6+ months experience building scalable backend services and AI-powered analytics platforms. Proven track record delivering production systems processing 100K+ records with sub-second response times. Expertise in LLM integration, natural language processing, multi-database architectures, and REST API development.

Technical Skills

Programming: Python (Advanced), SQL (Advanced), JavaScript (Intermediate), R (Basics)

Backend: FastAPI, Flask, REST APIs, Asynchronous Programming, Microservices

AI/ML: LangChain, Ollama (Llama3), TensorFlow, Keras, CNNs, NLP, Prompt Engineering

Data: Pandas, NumPy, pandasql, Feature Engineering, ETL Pipelines

Databases: MySQL (Multi-database), SQLite, NoSQL, Query Optimization

Tools: Docker, Git, Linux, Streamlit, Grafana, Prometheus, OpenCV

Experience

Seekright AI — *Python Backend & AI Engineer Intern*

Feb 2025 – Aug 2025

- Built scalable Python backend services supporting AI-driven workflows for 6+ organizations
- Developed LLM-powered SQL query assistant using LangChain and Ollama, reducing query development time by **80%**
- Implemented multi-database architecture with cross-database JOIN operations across organizational boundaries
- Created GPS data processing pipelines with coordinate conversion, improving spatial accuracy by **25%**
- Optimized backend throughput by **30%** through asynchronous programming and intelligent caching
- Built system monitoring dashboards using Prometheus and Grafana, reducing downtime by **15%**

Key Projects

DBLLM – AI-Powered SQL Analytics Assistant

Python, LangChain, Streamlit, MySQL

- Developed intelligent dashboard enabling natural language querying across 6 organizational databases with 100K+ records
- Implemented LLM-based text-to-SQL conversion with context-aware prompt engineering, achieving **95% query accuracy**
- Built secure multi-tenant backend with query validation, result caching, and pagination (100 records/page)
- Created interactive Streamlit interface with real-time filtering and data visualization using Plotly
- Achieved **80% reduction** in query execution time through optimized database connections and caching

GPS Video Metadata Extraction Tool

Python, OpenCV, Geospatial Processing

- Engineered automated tool extracting GPS coordinates, timestamps, and speed data from MP4 video files
- Implemented DMS-to-decimal coordinate conversion algorithms with **99.9% accuracy**
- Developed GPS interpolation algorithms improving data completeness by **35%**
- Optimized processing to handle 1GB+ video files in under 5 minutes with structured CSV output

Rice Disease Detection System

TensorFlow, CNN, Computer Vision

- Developed CNN-based image classification model achieving **94% accuracy** on rice disease detection
- Implemented data augmentation and transfer learning techniques for improved model generalization
- Optimized model using TensorFlow Lite for deployment, reducing model size by **75%**

Education

Bachelor of Computer Applications (BCA) — Vignanasudha Degree College

2021 – 2024

Relevant Coursework: Data Structures & Algorithms, Database Management, Software Engineering, Machine Learning

Certifications & Achievements

• **Python Advanced Programming** — HackerRank

May 2025

• **SQL Database Management** — MySQL

June 2025